MATERIALS SCIENCE

Hydroplastic foaming of graphene aerogels and artificially intelligent tactile sensors

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Direct foaming from solids is the most efficient method to fabricate porous materials. However, the ideal foaming fails to prepare aerogel of nanoparticles because the plasticity of their solids is denied by the overwhelming interface interactions. Here, we invent a hydroplastic foaming method to directly convert graphene oxide solids into aerogel bulks and microarrays, replacing the prevalent freezing method. The water intercalation plasticizes graphene oxide solids and enables direct foaming instead of catastrophic fragmentation. The bubble formation follows a general crystallization rule and allows nanometer-precision control of cellular wall thickness down to 8 nm. Bubble clustering generates hyperboloid structures with seamless basal connection and renders graphene aerogels with ultrarobust mechanical stability against extreme deformations. We exploit graphene aerogel to fabricate tactile microarray sensors with ultrasensitivity and ultrastability, achieving a high accuracy (80%) in artificially intelligent touch identification that outperforms human fingers (30%).

INTRODUCTION

Ultralight aerogel with high porosity is an important macroscopic form to harness the extraordinary surface properties of nanomaterials and has exhibited great applicable potentials in energy storage, catalysis, thermal insulators, sensors, and composites (*1*–*5*). In the example of graphene aerogel (GA), two basic preparation approaches have been developed: sol-gel and templating methods. Both methods start from dilute dispersions of solvable graphene or its derivatives (*6*, *7*). In the conventional Kistler's sol-gel method (*8*), dispersed graphene derivatives form interconnected gel and then specific drying techniques, such as supercritical drying (*6*) and freeze-drying (*7*), are usually adopted to prepare GAs to avoid structural collapse. This sol-gel method has been extended to prepare aerogels of many nanoparticles with different topologies, such as MXene (*9*) and carbon nanotubes (*10*), but limited by the high cost of specific drying and the imprecise regulation of porous structure. For the other templating method, sacrificial templates (*11*–*13*) are introduced to support graphene, and the removal of templates leaves graphene to form porous networks, represented by the freeze-dried (FD) GA with ice as template. Despite the high designability in structure and functionalities, the burdened removal of templates limits the scalable production of GA, and the cracking of ice or template boundaries (*14*) inevitably generate structural defects, resulting in poor structural integrity to resist complicated deformations. To date, the prevalent freeze-drying method has been widely adopted for small-scale preparation, yet unfulfilled the authentic industrial production with high efficiency as thermal plastic polymeric sponges.

Direct foaming from solids has become a possibly ideal method to prepare highly porous materials, exemplified by the thermoplastic foaming for various polymeric sponges (*15*), including polystyrene and polyurethane (PU). Distinct from the freeze-drying method, this "top-down" foaming from solid eliminates the burden of high energy costs of specific drying and the necessity of templates and prevents freeze-cracking, enabling millions of tons of polymeric sponges every year. The prerequisite of direct solid foaming is the high enough mobility of constituent molecules to adapt the bubble formation to form continuous connections without catastrophic breakage (*15*). This plastic state of solid is commonplace for polymers, but not for solids of two-dimensional (2D) graphene sheets, in which the overwhelming interlayer interactions immobilize graphene sheets and defy their plastic state. Previous reports have tried the thermal expanding method to prepare small-size GA samples (*16*) but usually obtained fluffy powders without any structural integrity. To date, realizing the direct solid foaming GA as industrial polymeric sponges remains a great challenge.

Here, we invent a hydroplastic foaming (HPF) method to directly convert laminated graphene oxide (GO) solid into GAs, not only in continuous bulks but also in large-area microarrays. HPF achieves the direct foaming of GAs similar to industrial thermoplastic polymeric sponges. The intercalation of water into GO interlayer space provides the plasticity for bubble nucleation and growth. We reveal that the bubble formation follows the classical crystallization rule, allowing the precise control of the cellular wall thickness (8 to 40 nm) and density (5 to 20 mg cm^{-3}) of GAs. The stable bubble clustering generates seamless basal connection of 2D sheets and renders GAs with ultrarobust mechanical stability against extreme deformations, such as tear and shear, comparable to polymeric sponges. Directly foamed GA simultaneously performs a high gauge factor (GF) (~2) and an ultrabroad strain range (0 to 95%) with ultrastability (for 10^4 cycles), outperforming most strain sensors. By integrating the direct HPF with ink printing, we fabricate large-area GA microarray sensors with a spatial accuracy of ~100 μm. In the frame of deep learning, the flexible GA arrays are used as an artificially intelligent (AI) tactile sensor to achieve over 80% accuracy in identifying material species and surface structures, far exceeding the average capability of human fingers (30%).

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RESULTS AND DISCUSSION

Fabrication and mechanism of HPF GAs

Direct foaming GO solids were continuously processed by three steps (Fig. 1A), which are water plasticization, bubble foaming, and direct drying. In the first hydroplastic step, the water intercalation turns stiff GO solid (fig. S1A) into a plastic state. As GO solid films contacted water, water molecules penetrated the GO interlayer gallery and the intercalation was completed in a few seconds (Fig. 1B) (*17*), driven by its hydrophilicity (fig. S1, B and C). The interlayer spacing increased from 0.90 nm in dried GO to 2.32 nm in plastic GO solid (GO/water = 1/3, volume ratio), revealed by the 2θ peak shifting from 9.8° to 3.2° in x-ray diffraction spectra (fig. S2A). An intuitional softening phenomenon was observed where the plasticized GO films can recover to original states without fracture after multiple folding (fig. S2B). Mechanical tests showed that the interlayer peel strength decreases by ~12.7 times after water intercalation (fig. S2C) owing to the considerably weakened interlayer interaction. The decline of elastic modulus for the hydroplastic GO film (fig. S2D) demonstrates the enhanced mobility of GO sheets. We defined this process as hydroplasticization in contrast with the thermal plasticity of polymers.

In the following bubble foaming, the reaction of N_2H_4 with oxygen-containing groups of GO sheets or the decomposition of other foaming reagents ($NaBH_4$ and $NaHCO_3$) exhausted gas (*18*) (such as CO_2, CO, H_2, and H_2O) and generated bubbles in the interlayer space (fig. S3 and movies S1 and S2). According to the Henry gas theory (*19*), the decreasing interlayer attraction in plastic GO weakens the constraint of bubble growth and lowers the bubble nucleation threshold pressure, facilitating the gradual foaming (fig. S4). The activated mobility of GO sheets in the hydroplastic state renders their rearrangement by sheet slippage and deformation to adapt the bubble growth (Fig. 1C). This foaming process resembles the thermoplastic foaming of polymeric sponges, in which plasticity is just triggered by thermal activation of polymer chains (*15*). In contrast, direct foaming GO solids without plasticization treatment only yielded high-density porous film (fig. S5) (*18*) or even fluffy powders (*16*), caused by the catastrophic expansion of GO laminates.

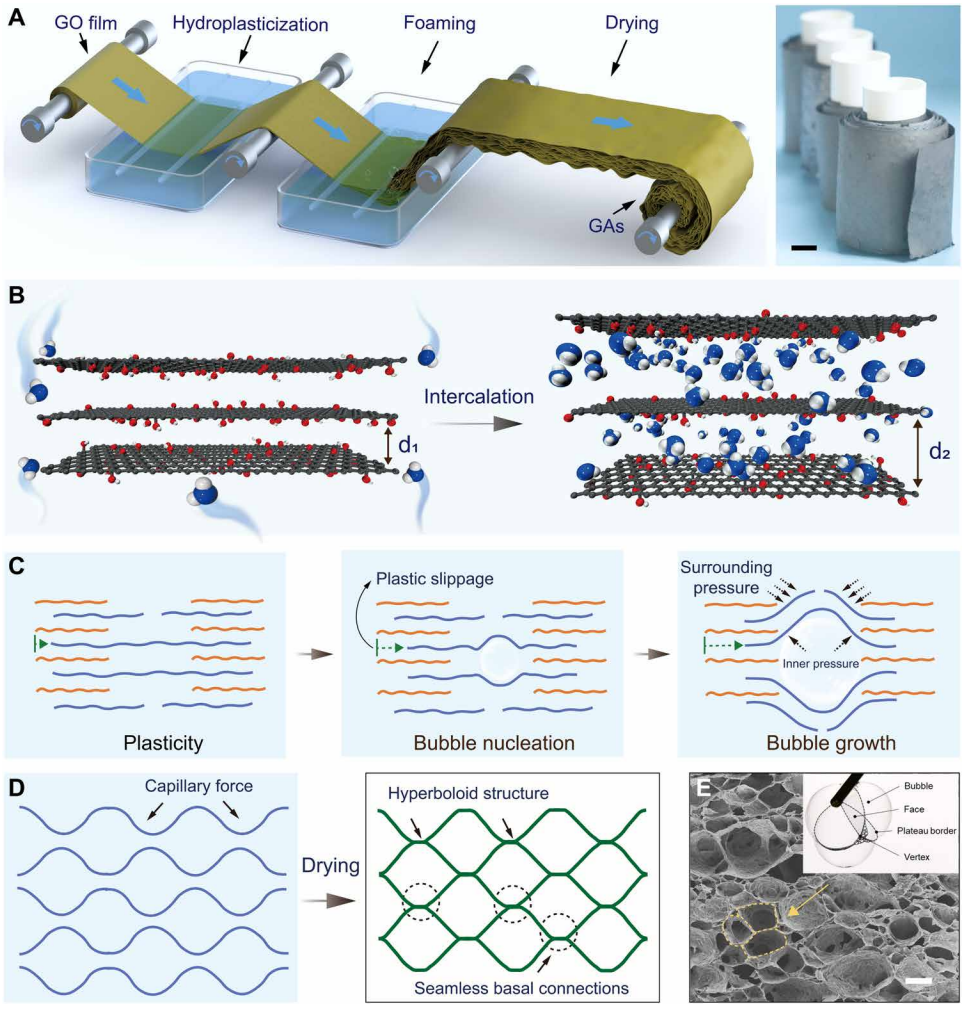


Fig. 1. Fabrication and mechanism of HPF GAs. (**A**) The process of HPF. GO solid film was continuously transformed to rolled GAs (~10 m, the photo in right panel) via three steps: hydroplasticization, bubble foaming, and direct drying. (**B**) The intercalation of water into GO solid to expand the interlayer spacing. (**C**) Self-structural adjustment of GO layers through plastic slippage to conform to the bubbling process to maintain structural integrity and avoid catastrophic expanding burst. (**D**) Seamless basal connections of graphene sheets formed by capillary-driven bubble clustering after drying. (**E**) SEM images of dried GA and the corresponding structural model of three-bubble connection (inset). Scale bars, 3 cm (A) and 200 μm (E). Photo credit: Kai Pang, Zhejiang University.

In the drying process, water was directly evaporated by heating to avoid unnecessary freeze-drying (*7*) and solvent replacement (*20*). During drying, the capillary force drives the clustering of bubbles and the adaptable GO walls bind together to form a continuous framework of GAs (Fig. 1D), obeying Plateau's law to minimize the surface area (*21*). The geometry of connected GO walls of aerogel has similar attributes to the clusters of soap bubbles, such as spherical walls, faces, Plateau borders, and vortex voids, as demonstrated in a three-bubble contact model (Fig. 1E). In HPF GAs, the face-to-face configuration of graphene walls dominates to form a hyperboloid structure (positive Gaussian curvature) with seamless basal connection, instead of staggering configurations in FD aerogels (fig. S6). The plastic foaming proceeds continuously and homogeneously even for continuous GA rolls (length of 10 m) and large monoliths (size of 30 cm by 15 cm) (Fig. 1A and fig. S7), eliminating the inhomogeneous volume expansion and stress cracking in the prevalent freeze-drying method. Moreover, the HPF method can possibly be extended to almost any nanoparticles, and we have specifically shown that it works for 1D cellulose nanofibers (fig. S8).

Bubbling mechanism and structural control

The HPF method exhibits high controllability on the microstructure and porosity (or density) of GAs, by controlling the nucleation density (N_n) and the size of bubbles (R_b). Figure 2 (A and B) shows the conclusive diagram of N_n and R_b, revealed by real-time optical microscope observations in the ultrathin GO film model (fig. S9 and movie S3). The bubble evolution can be described as two separate elementary processes in the classic crystallization theory, which are nucleation and growth. N_n is regulated by the concentration of the foaming reagent N_2H_4 (C_{N2H4}) and increases from 162 to 794 mm^{-2} as C_{N2H4} increases from 10 to 85% [weight % (wt %)] at room temperature, conforming to a linear function of $N_n = 12.4 + 8.68C_{N2H4}$ (fig. S10A). The nucleation rate approaches an equilibrium with C_{N2H4} increasing to 50% (fig. S11A). Raising temperature accelerated the gas exhaust and therefore the nucleation rate of bubbles (fig. S11B). At a given N_n, the bubble growth follows the Ostwald ripening rule (*15*), and R_b increases from 6 to 470 μm as the HPF time (t, min) extends, obeying a relationship of $R_b = 15.3\ t^{-0.8}$ (fig. S10B).

The clarified evolution of bubble nucleation and growth allows the precise control of the microstructure and porosity of GAs. Bubbles act as separators to delaminate thick GO laminates into thinner walls and determine the pore size (R_p), which can be described by a sinusoid wavy surface model (fig. S12). Following this top-down principle, higher N_n generated thinner walls (Fig. 2D) and the wall thickness (T) was precisely tuned by controlling N_n following an exponential relationship of $T = 33N_n^{-1.0}$, as shown in the statistical results by scanning electron microscopy (SEM) measurements (Fig. 2C and fig. S13, A and B). This exponential relation equates the theoretic scaling law of $T \sim N_n^{-1}$ (fig. S12). High-resolution transmission electron microscopy (HR-TEM) analysis revealed that the wall thickness was tuned from ~40 to ~8 nm, reaching a high accuracy in the nanometer scale. The porosity, or the apparent density (ρ), is tuned by R_b and t, at given N_n and temperature. The average R_p of GAs expanded from 40 μm at 0.17 min to 190 μm at 120 min (fig. S13C). As a result, ρ decreased with the extending t and expanding of R_p, obeying the power relations of $\rho = 13.3\ t^{-0.2}$ and $\rho = 386.4\ R_p^{-0.8}$ (Fig. 2E and fig. S10C), close to the theoretical relation of $\rho \sim R_p^{-1}$ (fig. S12). The control on T and R_p can be separated by tuning N_n and t, respectively (fig. S13, E and F). At a given N_n, the wall thickness of GAs at different times and temperatures barely changed (fig. S14, A to E). The benefits of HPF methodology include not only high productivity and low costs by avoiding complex drying processes but also the precise yet facile controllability on microstructures of GAs by the top-down bubbling strategy, which are difficult for conventional sol-gel and templated methods (table S1).

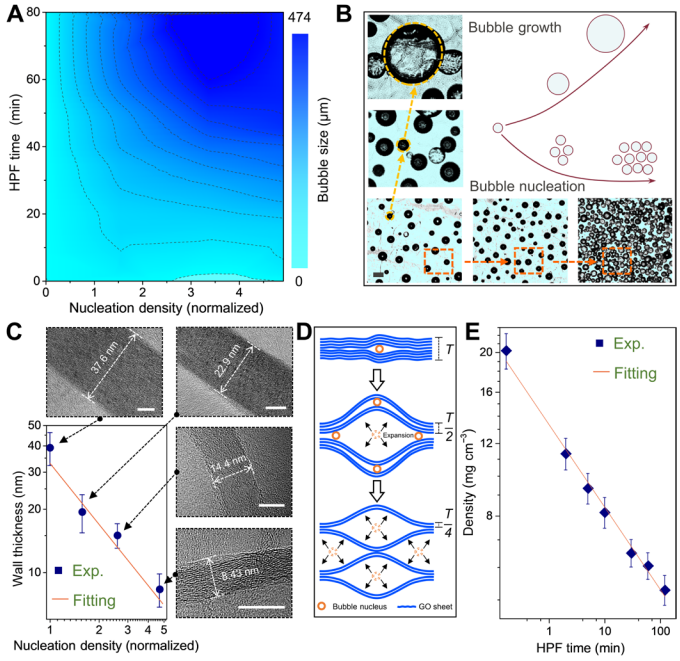


Fig. 2. Bubbling mechanism and structural control. (**A**) A conclusive bubble diagram of N_n and R_b by in situ optical microscope observation of the HPF process at room temperature with different C_{N2H4}. (**B**) Snapshots of bubble nucleation (increasing density) and bubble growth (bubble expanding) and their corresponding diagrams. (**C**) The tunable wall thickness of GAs by N_n and corresponding HR-TEM images. The fitting curve is $T = 33N_n^{-1.0}$. (**D**) Schematic of decreasing wall thickness with bubble nucleation density. (**E**) Numerical correlation of HPF time with density of GAs ($\rho = 13.3\ t^{-0.2}$) at 60°C and C_{N2H4} of 85 wt %. Scale bars, 50 μm (B) and 10 nm (C).

Mechanical robustness of HPF GAs

The hyperboloid structures with the seamless basal connection of cellular graphene walls and the structural integrity enable HPF GAs to endure complex deformations, including the ordinary compression and the extreme tearing and shearing. To assess the structural integrity under extreme deformations, the bulk HPF GA (ρ of 3.5 mg cm^{-3}) was crumpled to its 30% size and forced to pass through a smaller pipe without any rupture (Fig. 3A and movie S4). The global crumpling has complicated deformation patterns and includes compression, stretching, tearing, and twisting. The survival of HPF GAs under this harsh test denotes the superior mechanical robustness of polymeric sponges, which has never been achieved in the neat FD GAs (fig. S15).

To quantitatively evaluate mechanical robustness, we measured HPF GAs in different elementary deformation modes: compression, tension, bending, shearing, and tearing, in comparison with the traditional FD aerogel. HPF GA bulks exhibited excellent resilience at 90% compressive strain for 10^5 cycles and kept a small variation of

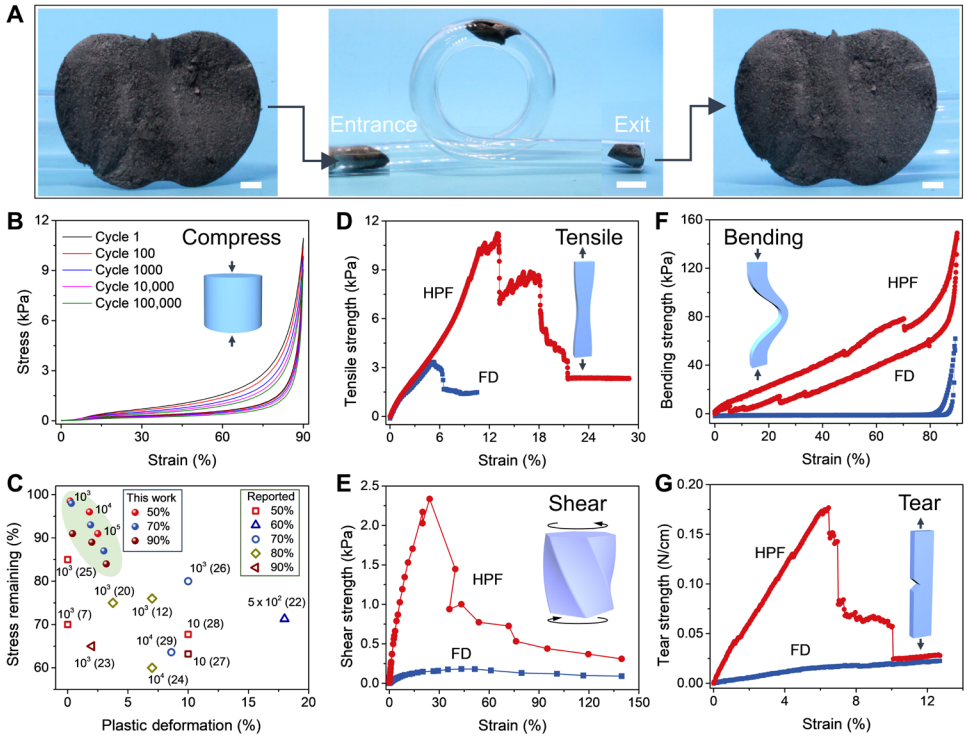


Fig. 3. Superior mechanical robustness of HPF GAs. (**A**) HPF GAs survive the extreme deformation test by passing through a smaller pipe, exhibiting superior structural integrity without any breakage. (**B**) Stress-strain curves of GAs under high compression strain (90%) for 10^5 cycles. (**C**) A comparison chart plotting plastic deformation versus stress remaining during different compression cycles for HPF GAs and previously reported GAs. (**D** to **G**) Mechanical tests under elementary deformation modes: tensile (D) shear (E), bending (F), and tear (G) tests. Scale bars in (A), 10 mm (left and right) and 20 mm (middle). Photo credit: Kai Pang, Zhejiang University.

energy loss coefficient of ~0.05 (Fig. 3B and fig. S16A). Under long-term compression for 15 days at 99% strain, HPF GAs held a high recovery (99% stress and 100% strain remaining) with negligible relaxation (fig. S16B). Previous studies on FD GAs (*7*, *11*, *12*, *20*, *22*–*29*) only exhibited sound, pure-compression resilience; however, our HPF GAs elevated the capability limit on the minimal plastic deformation and the remaining maximal stress (Fig. 3C). The compression resilience of HPF GAs remained stable in a wide temperature range from −196°C (in liquid nitrogen) to 400°C, with a 100% retention of the maximum stress (fig. S16C). In addition, the Young's modulus (E) of HPF GA correlated with density as $E \sim \rho^{0.83}$ (fig. S21, A and B), and thinner walls were beneficial for the improvement of compressive strength and modulus (fig. S21, C and D).

Mechanical tests in other elementary deformation modes demonstrated that HPF GAs behave stronger and tougher than FD GAs with the same ρ of 5 mg cm^{-3} (Fig. 3, D to G). In tensile tests, HPF GAs exhibited 340% higher fracture strength and 250% higher breakage elongation than that of FD aerogels (Fig. 3D). The bending modulus and strength are 11- and 2.43-fold higher than those of FD aerogels, respectively (Fig. 3F). The remarkable superiority of HPF GAs is the strength against shear and tear, outperforming FD aerogels by 12.4 (Fig. 3E) and 10.4 times (Fig. 3G), respectively. Previous reports (*7*, *11*, *12*, *20*, *22*–*29*) and our control experiments implied that FD aerogels could not endure the extreme shear and tear deformations because of freeze-cracking and weak interconnection. As a contrast, HPF GAs even exhibited remarkable recoverable abilities to tensile, bending, and shear tests for more than 1000 times, similar to polymeric sponges (fig. S17). The overall superior mechanical performances of HPF GAs originate from the hyperboloid structure with seamless connection and the absence of cracks. In situ SEM inspections revealed that the basal graphene walls could adapt deformations without breakage, but the point-to-point and point-to-face connections between cellular walls in FD aerogels were weak points to break easily (fig. S18).

High-sensitive GA strain sensor

The high compressibility and superior mechanical robustness of HPF GAs guarantee the prerequisite of stability and reliability as strain sensors based on the piezoresistive effect. We assessed the sensor merits of a single HPF GA sensor that was encapsulated on silver electrodes by PU film (Fig. 4A), and it exhibited high sensibility in a wide strain range (fig. S19A). The GF, an important parameter to evaluate the response sensitivity, exhibits three linear regimes: GF = 2.01 in the strain range of 0 to 7%, GF = 0.55 in 7 to 70%, and GF = 1.24 in 70 to 95%. The GF and sensing curves have a small fluctuation for 10^4 test cycles at a high strain of 90% (Fig. 4C and fig. S22A). The evolution of resistance fits a single monotonic function of $\eta = -0.0004\,\varepsilon^2 + 0.016\,\varepsilon + 0.02$ (where η is the relative resistance change $\Delta R/R_0$ and ε is the mechanical strain), which can be used as a master curve for reliable data processing in practical applications (Fig. 4B).

For most strain sensors, the strain sensitivity and sensing strain are a pair of paradoxes (*12*, *30*–*34*). The HPF GA sensor with high porosity (above 99%) fills the gap by providing an ultrabroad strain range and a reliably high GF, simultaneously. On one hand, given the high GF, the sensing range (0 to 95%) of the HPF GA sensor is broader than that of (usually below 5%) conventional piezoresistive sensors based on conductive networks of nanowires (*30*), microcracking

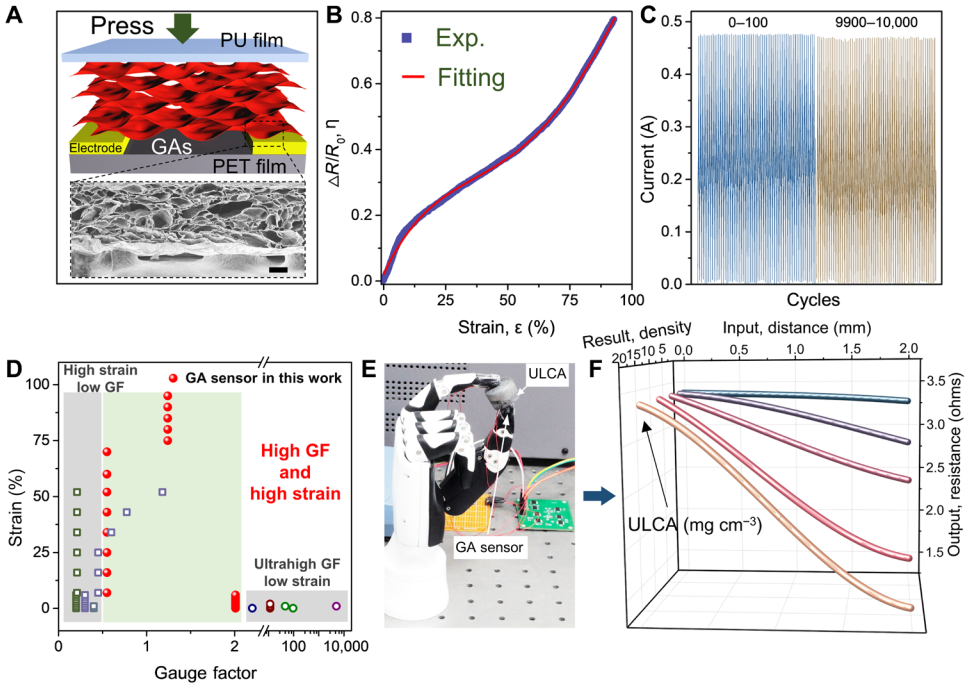


Fig. 4. High sensitivity of the GA sensor. (**A**) Structure of the GA press sensor and its SEM image. (**B**) The resistance change ratio ($\Delta R/R_0$, η) of the GA sensor to 95% strain and its fitting as $\eta = -0.0004\varepsilon^2 + 0.016\varepsilon + 0.02$. (**C**) Fatigue resistance of the GA sensor up to 90% strain for 10^4 cycles. (**D**) The comparison between the GA device and other piezoresistive sensors (*30*–*32*) and carbon aerogels (*12*, *33*, *34*), in the merits of GF and strain range. (**E**) Mechanical control of the robotic hand with the GA sensor to identify the density of ultralight carbon aerogel (ULCA). The responsive data signals were collected by an eight-channel data acquisition card. (**F**) The cubic fitting curves between the press distance and resistance to determine the ULCA density by a Gaussian kernel process (fig. S20 and movie S5). Scale bar in (A), 50 μm. Photo credit: Kai Pang, Zhejiang University.

(*31*), and spacing regulation (MXene film) (*32*). On the other hand, the GF of HPF GA behaves much higher than other carbon aerogels, including carbonaceous nanofibrous aerogel (*33*), carbon nanotube aerogel (*34*), and biomimetic GA (*12*) prepared from conventional sol-gel and templating methods (Fig. 4D). Moreover, the lowered density or wall thickness of GA enhanced GF at small strains (fig. S21, D and E). Besides the high GF and broad sensing range, the GA sensor exhibited an exceptionally high sensitivity of force down to 4 mN and pressure to 25 Pa (fig. S19, B to D), benefiting from its high porosity for easy deformation (Fig. 3B). For the extreme circumstance, the GA sensor is able to recognize the density change of ultralight carbon aerogels, even down to 1 mg cm^{-3}. We patched the GA sensor on the fingertips of a robotic hand to pinch ultralight carbon aerogels (Fig. 4E). The GA sensor identified the density difference of carbon aerogels by either slight touching (low strain) or hard pinching (high strain), judging from the response curves of resistance by the Gaussian kernel process (Fig. 4F, fig. S20, and movie S5) (*35*). Moreover, to further evaluate the reliability of GA sensors for practical applications, we have prepared 10 batches of GAs and compared their mechanical and sensing performances (fig. S22, B and C). We found that there was little change in their compressive and strain sensing curves, demonstrating the high repeatability and reproducibility of the HPF process.

AI microarray sensor of GAs

We developed a direct method to fabricate large-area GA sensor microarrays (Fig. 5A and fig. 23, A and B), by integrating the ink-printing technique with HPF. GO was ink-printed between electrode channels on polyethylene terephthalate (PET) substrate and the HPF converted deposited GO arrays to GA arrays, followed by chemical reduction and PU capsulation. The area can be easily scaled up to reach a continuity to match continuous printing. We were able to achieve the GA unit with a size resolution of 100 μm (fig. S23C), much smaller than the ridge width of the adult human fingerprint (~500 μm). Large-area GA sensor arrays (11 cm by 16 cm, 100 pixels) can map the shape of objects and compression distribution on them (Fig. 5B), such as a hand contour in Fig. 5A (see details in fig. S23D).

We equipped the GA microarray sensor (64 sensors on 7 mm by 8 mm; Fig. 5C) onto a robotic fingertip and evaluated its sensing ability to identify micro-sculpture patterns (a model of English letters A to Z) (Fig. 5D), aided by the deep-learning data processing. Testing signals (fig. S24B) were acquired by signal modification circuits (Fig. 5E and fig. S24A) and then converted into pixels by normalizing to the range of 0 to 255 and assembled as an 8 × 8 letter image corresponding to their locations (Fig. 5F). Collected letter images were enhanced 64 times and used as test inputs into the trained convolutional neural network (CNN) model, a usual model for sensor data analysis (*36*–*37*), to conduct practical letter identification (see movie S6). More than 10^5 letters in Eminist datasets (*38*) were used to train the CNN by back-propagating errors (fig. S25A). The output confusion matrix of actual and predicted object labels (fig. S25B) indicated the considerable variation of classification accuracy for individual letters due to the image variety in the handwriting Eminist dataset.

The ultrastability of the GA array sensor enables the reliability of massive data (fig. S26), setting the basis for reliable AI tactile finger

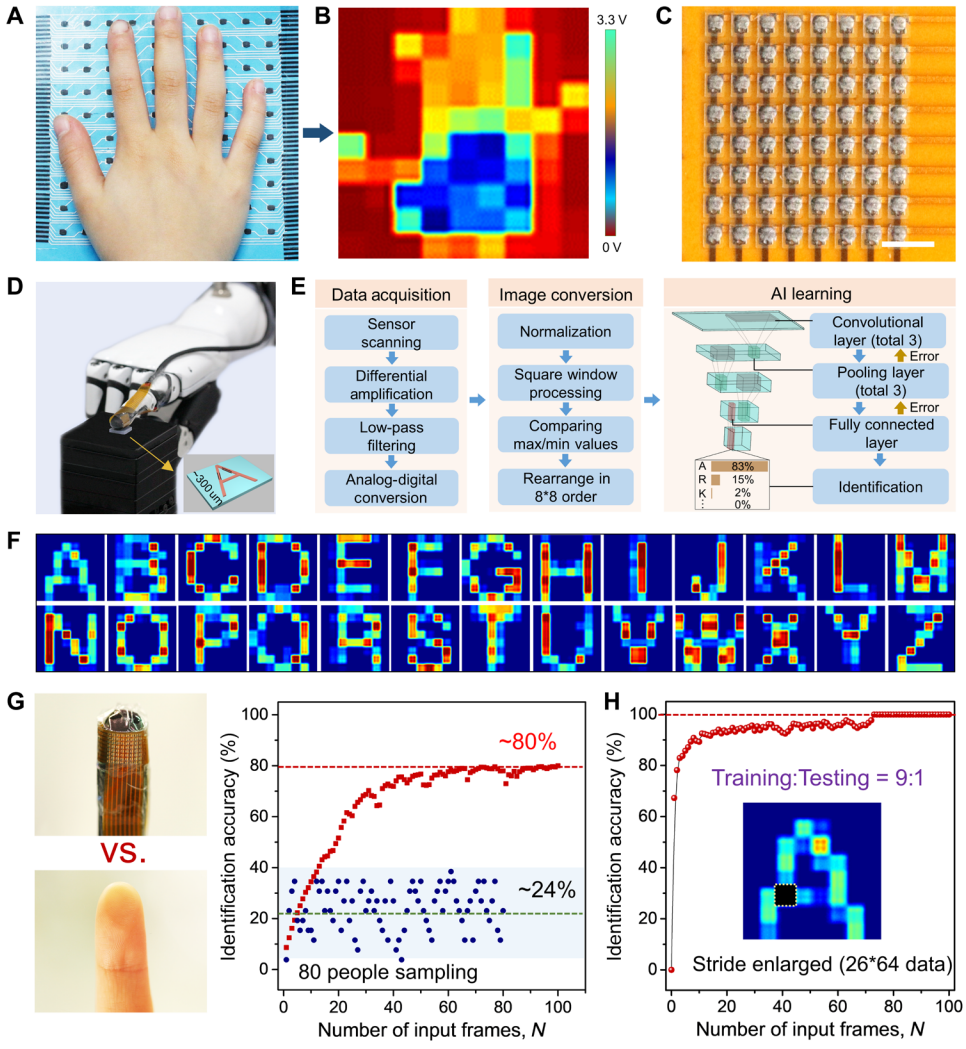


Fig. 5. AI microarray sensor of GAs. (**A**) Hand pressing on a 10 × 10 GA array sensor (single pixel, 0.5 mm by 0.7 mm). (**B**) Pressure distribution maps of hand [in (A)]. (**C**) Photo of an 8 × 8 microarray sensor with a single cell size of ~300 μm. (**D**) A robotic finger with a GA microarray sensor is touching the letter patterns. (**E**) Flowchart of AI identification, including data acquisition, image conversion, and AI learning. The trained CNN by error back-propagation with the Eminist dataset (over 10^5 letters) as training set contains three convolutional layers, three pooling layers, and two fully connected layers. (**F**) Collected letter images of A to Z through flow [in (E)]. (**G**) The identification contest between an AI GA finger sensor and an adult human finger. The average accuracy of the finger sensor by using a diverse set of tactile maps with *N* distinct clusters as input and the accuracy of the human finger by sampling from 80 people. (**H**) The letter identification accuracy is close to 100% by using collected letter images with stride enlarged technics (64 times for 26 letters) as a training dataset. Scale bar in (C), 1.5 mm. Photo credit: Kai Pang, Zhejiang University.

sensors. The accuracy of the AI GA tactile sensor to identify letters reached 80% (Fig. 5G), using a diverse set of tactile maps (from A to Z) with inputting *N* distinct clusters (1 to 100). We launched a competitive test between an AI GA finger sensor and an adult person's finger in the letter identification (Fig. 5G). The average accuracy of the GA finger sensor (80%) was overwhelmingly higher than that (30%) of human fingers (80 people samples), simultaneously performed at a faster response speed (within 1 s). Moreover, a stride enlarging method was conducted by adding a black pixel for 64 times in the acquired images to simulate the broken situations. The enlarged datasets (26 × 64 images) were divided into training and testing classes (9:1) to test the probability of self-learning, and the GA microarray sensor reached the perfect accuracy of letter identification (near 100%) (Fig. 5H). Through machine intelligence training, the accuracy of the finger sensor with eight GA arrays to identify the material species can be gradually enhanced to nearly 88%. Seven types of materials were identified, including stainless steel, polylactic acid, wood, hardboard, silicon rubber, sponge, and tissue (see details in fig. S27). The identification accuracy of our GA tactile sensor has reached a leading level of wearable sensors in recognition of materials species and pattern (*36*, *37*).

Conclusions

We present a new HPF method to fabricate ultralight aerogels of nanomaterials directly from their solid, mainly exemplified by GA. The underlying hydroplastic and bubbling mechanisms are revealed to guide the precise regulation of cellular wall thickness and porosity of aerogels, extending our capability to design and control structures of porous materials. Following the HPF mechanism, GAs intrinsically feature seamless basal connection and structural

homogeneity in large scale, determining their superior mechanical robustness to endure complex and extreme deformations, greatly relieving the structural weakness and brittleness of GAs with ultrahigh porosity. Beyond the pure compressibility in previous reports, hydroplastic foamed GA exhibits an all-around structural integration against complex deformations, performing like the daily used polymer sponges.

Extending from the ordinary aerogel bulks, we integrate the HPF method with an ink-printing technique to continuously produce GA microarrays with high size precision in a large scale. The GA strain sensor exhibits both high sensibility and wide strain range, filling the gap between conventional piezoresistive sensors and other conductive porous sensors. The superior dynamic structural stability of GA frameworks guarantees the high sensing stability and massive reliable data for the use of AI tactile sensor devices. The AI GA tactile sensor performs a much higher accuracy exceeding 80% to identify material species and micro-sculpture patterns, showing a complete superiority to human fingers. As a versatile methodology, our HPF promotes the industrial production of aerogels of graphene and other nanomaterials, replacing the prevalent freeze-drying method. The greatly enhanced mechanical robustness increases the practical worth of aerogel materials for broad applications in advanced nanocomposites, smart electronics, thermal management, absorption, and separation.

MATERIALS AND METHODS

Fabrication of GO films

Aqueous GO dispersions were purchased from Hangzhou Gaoxi Technology Co. Ltd. (www.gaoxitech.com). Typically, aqueous GO suspension (8 mg ml^{-1}) was cast-dried on the PET substrate and peeled off to obtain freestanding GO films.

Direct fabrication of GAs

For the direct HPF process, the GO films were immersed into the N_2H_4/H_2O solution. The N_2H_4 ratios (10, 30, 55, and 85 wt %), temperature (20°, 40°, 60°, 80°, and 100°C), and time were tuned to control the nucleation density, foaming rate, and pore size, respectively. The GAs were obtained after drying. The GAs for mechanical tests and sensor evaluation were further annealed at 1600°C.

Continuous fabrication of GAs

The rolled GO films were pulled into the water and aqueous solution of foaming agents (30% N_2H_4 solution, wt/wt); thus, the continuous GAs were achieved after air-drying at 60°C.

In situ HPF fabrication of GA array

Solid GO microunits were obtained by the direct printing with high-concentration GO suspension (30 mg ml^{-1}) on the designed electrical circuits. GA array was achieved by in situ squeezing of N_2H_4/H_2O solution (30%, wt/wt) on the GO solid for 10 min to proceed with the HPF process. The electrical conductivity of GAs for piezoresistive sensing was further enhanced (9.7 ± 2 s/m) by hydroiodic acid (HI) reduction at 90°C. The wall thickness and density of the GA in the sensor array were about 20 ± 4 nm and 25 ± 2 mg/cm^3, respectively.

Device of GA array sensor for AI identification

The sensor array for letter identification was grown on a flexible print circuit (FPC) fabricated by the zedoary process. The size of the FPC was 7 mm (L) by 8 mm (W) by 0.13 mm (H) with a total of 64 electrode pairs, where a single pair was shaped as 0.65 mm (L) by 0.5 mm (W). Sensor signals were obtained by scanning reading, and its circuit topology was conducted by buried vias on electrodes. The GA sensor units were fabricated by in situ HPF processes. This tactile array of GA sensors was attached on the robotic finger by the PU encapsulation with superior flexibility and stability.

Machine learning and identification

The machine learning for letter identification was to classify the sensor image. Raw image inputs were unified by comparing the differences between activation values and initial values of sensor outputs. The generated image was scaled up to 64 × 64 to fit the input standard of the later mentioned CNN. A modified LeNet-5 was adopted to conduct the classification task to gain an optimal result. It consisted of three convolutional layers, three maximum pooling layers, two fully connected layers, and one softmax output layer. It was trained on a deep learning platform with 32 GB random-access memory, dual 2080Ti graphics processing unit (GPU), and Core i9 9900k central processing unit (CPU). Furthermore, the size of the convolution kernel was 5 × 5, and the convolution stride was 2. The loss function of backpropagation was optimized by the Adam algorithm. To avoid overfitting, the dropout function was used in the networks.

The Eminist dataset was selected as the training dataset of this work to verify the model generalization and ensure its robustness. The Eminist dataset was also reshaped as 64 × 64 to meet the input requirement of the abovementioned neural networks, and the correction rate of this task achieved 80%. Moreover, the classification was conducted by using the sensor images as both training and testing data with a proportion of 9:1.

Characterizations

The structure and morphology were investigated by optical microscopy, SEM, HR-SEM, HR-TEM, and atomic force microscopy on ZEISS Axio Scope, Hitachi S4800, ZEISS Utral 55, JEM-2100, and VEECO Multimode systems, respectively. To determine the accurate wall thickness by HR-TEM, the GAs were embedded with epoxy resin and then sliced ultrathin for further observation. The homemade stage with manual step length was used to observe the structure of GAs during compression and bending. In situ SEM observation of GAs during compression and bending was manipulated by scanning at different strain states with a retractable sample holder.

The compressive tests were performed on a microcomputer control electronic universal testing machine (RGWT-4000-20, REGER). The Instron Legend 2344 machine is used for tensile, tear, and peel tests. Shear tests were taken on the HAAKE RS6000 machine. The change of electrical properties of GA sensors was evaluated by the combination of mechanical testing systems and the Keithley 2400 Source Meter.

SUPPLEMENTARY MATERIALS

Supplementary material for this article is available at http://advances.sciencemag.org/cgi/content/full/6/46/eabd4045/DC1

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